

Centrifugal Pump P-101

Operation and Maintenance Manual — Crude Charge Service — Document PM-P101 Rev. 4

SYNTHETIC DOCUMENT - created for a Smart India Hackathon prototype demonstration. Not real plant data. No sponsor information is used.

1. Equipment Data

Parameter	Value
Tag number	P-101 / P-101A (standby)
Service	Crude charge to atmospheric distillation
Type	Horizontal centrifugal, single stage, API 610 OH2
Rated flow	420 m3/h
Rated head	185 m
Design pressure	28 bar g
Normal suction pressure	2.1 bar g
Operating temperature	185 deg C
Driver	Induction motor, 315 kW, 2970 rpm

2. Mechanical Seal

P-101 is fitted with a single cartridge mechanical seal to API 682 Category 2, Arrangement 1, with Plan 32 external flush. The seal faces are silicon carbide against carbon.

The seal flush shall be maintained at 3 bar above suction pressure, measured at the flush line inlet.

Flush flow rate shall not fall below 8 litres per minute. Operation outside these limits will cause the seal faces to run dry and fail prematurely.

Flush liquid shall be filtered to 25 micron. A blocked or partially restricted flush line is the most common cause of repeat seal failure in this service, and should be the first item checked when the interval between failures is observed to be shortening.

3. Inspection Intervals

Item	Interval	Method
Mechanical seal assembly	6 months	Visual, leak check
Seal flush line and filter	3 months	Differential pressure
Bearing vibration	Monthly	Handheld analyser
Coupling alignment	12 months	Laser alignment
Casing wear ring clearance	36 months	Dismantle and measure

The mechanical seal inspection interval of six months applies to crude service. Where the interval between unplanned seal failures falls below 120 days, the cause shall be investigated before the seal is replaced again.

4. Start-up Procedure

1. Confirm suction and discharge valve positions.
2. Establish seal flush and confirm 3 bar differential before starting.
3. Vent casing at the high point until liquid appears.
4. Confirm standby pump P-101A is isolated.
5. Start driver and confirm discharge pressure within 10 percent of rated.
6. Log start time, suction pressure and flush differential in the shift log.

5. Fault Finding

Symptom	Likely cause	Action
Seal leakage within 90 days	Flush line restriction	Check filter dP, clean strainer
High bearing vibration	Misalignment or cavitation	Check alignment, confirm NPSH
Low discharge pressure	Wear ring clearance	Measure clearance at next shutdown
Seal face scoring	Solids in flush liquid	Verify 25 micron filtration